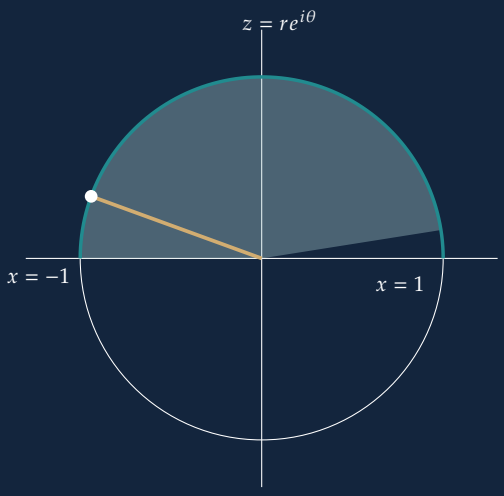
The Xi–Chebyshev Trace Identity

for the Riemann ξ-Function

~~Finite radius, global radial splitting, and QQRH~~

$$\int_{-1}^{1} \frac{\partial_x P(r,x)}{P(r,x)} U_{k-1}(x)\sqrt{1-x^2}\,\mathrm{d}x = \pi r^k q_k$$



October 2026 · Extended textbook edition

Two exact trace theorems, an angular anchor, and precise conditional RH criteria.

Preface

This second edition studies the finite-radius Chebyshev identity, derives a *global radial-split extension*, and explains the exact logical relationship between angular positivity and QQRH (a family of zero-exclusion wedges for the Riemann hypothesis, RH). The new global trace formula is valid at every positive radius in a logarithmic Fourier sense; its ordinary derivative form holds whenever the circle contains no zero. No RH assumption is used in either trace theorem.

The central idea is to change variables from the usual spectral coordinate s to

$$z = (s - \tfrac{1}{2})^2,$$

and to view the squared modulus of the centered entire Riemann function on circles $|z| = r$. A Chebyshev transform then recovers the inverse-power sums of the transformed zeros *exactly*. Those sums provide a spectral representation of the angular derivative whose positivity is used in QQRH.

The main distinctions are the following:

1. The original *finite-radius identity* follows from the genus-zero product of the actual completed zeta function.
2. A *global radial-split identity* remains exact after zeros enter the circle, with a logarithmic integral at critical radii.
3. The *finite-radius angular anchor* is a sign theorem, not merely an identity; it uses a separate zero-height input. Its value 255/7 has an exact factor-geometric derivation.
4. The *QQRH bridge* and the equivalence between RH and global full-angle positivity require sign control for unbounded radii. A spectral identity alone supplies no such sign.

A proof of the local and global trace identities is given in standard mathematical analysis. These proofs are *not* asserted to be newly formalized in Lean. The supplied Lean archive provides source-level context for the bounded-radius angular theorem; Appendix B records its precise provenance and limits.

Normalization to retain. If $q_k = -\sum_{\alpha \in \mathcal{Z}_\Theta} \alpha^{-k}$, counting zeros with multiplicity, the trace is $\pi\, r^k q_k$, not πq_k. Equivalently, one can define the radius-dependent coefficient $Q_k(r) = r^k q_k$ and write the trace as $\pi Q_k(r)$.

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CHAPTER 1

The centered entire function and angular profile

1.1 The Riemann function and the square coordinate

The completed Riemann function is

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s), \tag{1.1}$$

with its removable singularities filled in. It is entire, nonzero at $s = \frac{1}{2}$, real on the real axis, and satisfies

$$\xi(s) = \xi(1-s), \qquad \overline{\xi(s)} = \xi(\overline{s}). \tag{1.2}$$

Its zeros are precisely the nontrivial zeros of ζ, with multiplicity. Write

$$F(w) := \xi(\tfrac{1}{2} + w).$$

The functional equation gives $F(w) = F(-w)$, so F is an *even entire function*. Consequently there is a unique entire function Θ for which

$$F(w) = \Theta(w^2), \qquad \Theta(z) = \xi\left(\frac{1}{2} + \sqrt{z}\right). \tag{1.3}$$

The last expression is branch-independent: the two choices of $\sqrt{z}$ give the same value. Indeed, in the Taylor expansion $F(w) = \sum_{j\geq 0} c_{2j} w^{2j}$, we simply set $\Theta(z) = \sum_{j\geq 0} c_{2j} z^j$.

Definition 1.1 (Transformed zero multiset)**.** Let $\mathcal{Z}_\Theta$ consist of the nonzero zeros α of Θ, counted with analytic multiplicity. Each ξ-zero ρ is carried to

$$\alpha = (\rho - \tfrac{1}{2})^2. \tag{1.4}$$

The pair ρ and $1-\rho$ produces *one* zero α of the same multiplicity, not two. Complex conjugate ξ-zeros produce conjugate Θ-zeros.

1.2 Circle coordinates and the profile

Set $r > 0$, $0 \leq \theta \leq \pi$ and $x = \cos\theta$. The upper semicircle is

$$z_r(x) = r\left(x + i\sqrt{1-x^2}\right) = re^{i\theta}, \qquad -1 \leq x \leq 1. \tag{1.5}$$

The central objects are

$$P(r,x) = |\Theta(z_r(x))|^2, \quad A(r,x) = \partial_x P(r,x), \quad \mathcal{L}(r,x) = \frac{A(r,x)}{P(r,x)}. \tag{1.6}$$

We use $\mathcal{L}$ only where $P > 0$. This is the angular *logarithmic* derivative:

$$\mathcal{L}(r,x) = \partial_x \log P(r,x).$$

Although $z_r(x)$ contains a square root, the two conjugate arcs multiply to give a smooth, indeed real-analytic, function of x on the interior of $[-1, 1]$. Wherever the profile is nonzero, $\mathcal{L}$ is also real-analytic.

Two kinds of positivity. The elementary inequality $P(r, x) \geq 0$ is automatic from its definition and does *not* exclude zeros. The useful angular condition is $A(r, x) = \partial_x P(r, x) \geq 0$, or equivalently $\mathcal{L}(r, x) \geq 0$ when $P > 0$. It is a substantive monotonicity statement.

1.3 Chebyshev polynomials of the second kind

The polynomials U_n are defined by

$$U_n(\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}, \qquad 0 < \theta < \pi. \tag{1.7}$$

Thus $U_0(x) = 1$, $U_1(x) = 2x$, $U_2(x) = 4x^2 - 1$. Their orthogonality is

$$\int_{-1}^{1} U_j(x) U_k(x) \sqrt{1 - x^2}\, \mathrm{d}x = \frac{\pi}{2}\, \mathbf{1}_{\{j=k\}}, \qquad j, k \geq 0. \tag{1.8}$$

Proof: put $x = \cos\theta$, use $\sqrt{1 - x^2} = \sin\theta$, and reduce to $\int_0^\pi \sin((j+1)\theta)\sin((k+1)\theta)\, \mathrm{d}\theta$.

CHAPTER 2

Zeros, logarithms, and the exact trace

2.1 The canonical product in the squared coordinate

The classical order-one growth of ξ implies order at most $\frac{1}{2}$ for Θ: for large r, its maximum modulus grows no faster than $\exp(C\sqrt{r}\log(2+r))$. The genus-zero form of Hadamard factorization therefore applies:

$$\Theta(z) = \Theta(0) \prod_{\alpha \in \mathcal{Z}_\Theta} \left(1 - \frac{z}{\alpha}\right), \qquad \sum_{\alpha \in \mathcal{Z}_\Theta} \frac{1}{|\alpha|} < \infty. \tag{2.1}$$

The product is locally uniformly convergent, includes analytic multiplicities, and requires *no RH assumption*. Here and below $\Theta(0) = \xi(\frac{1}{2}) \neq 0$.

Let

$$R_0 := \inf_{\alpha \in \mathcal{Z}_\Theta} |\alpha| > 0. \tag{2.2}$$

For $|z| < R_0$, there are no zeros and there exists an analytic logarithm of Θ normalized at the origin. Logarithmic expansion of (2.1) gives

$$\log \Theta(z) = \log \Theta(0) + \sum_{\alpha \in \mathcal{Z}_\Theta} \log(1 - z/\alpha) \tag{2.3}$$

$$= \log \Theta(0) - \sum_{k=1}^{\infty} \frac{z^k}{k} \sum_{\alpha \in \mathcal{Z}_\Theta} \alpha^{-k}. \tag{2.4}$$

All sums can be interchanged normally on every smaller closed disk $|z| \leq r < R_0$.

Definition 2.1 (Inverse-power trace coefficients)**.** For integers $k \geq 1$, define

$$\boxed{q_k := -\sum_{\alpha \in \mathcal{Z}_\Theta} \alpha^{-k} \in \mathbb{R}.} \tag{2.5}$$

They are real because $\mathcal{Z}_\Theta$ is invariant under complex conjugation, with matching multiplicities. Hence

$$\boxed{\log \Theta(z) = \log \Theta(0) + \sum_{k=1}^{\infty} \frac{q_k}{k} z^k, \quad |z| < R_0.} \tag{2.6}$$

Remark 2.2. q_k is a spectral power sum associated with the *squared-coordinate* zeros of ξ. It is not, by definition, a Li coefficient. On RH, where every $\alpha = -\gamma^2$, formula (2.5) specializes to

$$q_k = (-1)^{k+1} \sum_{\gamma > 0} \frac{m_\gamma}{\gamma^{2k}},$$

with positive ordinates γ and zero multiplicities m_γ. This specialization must not be inserted into an unconditional proof.

2.2 Main theorem: the Chebyshev integral

Theorem 2.3 (Exact finite-radius Chebyshev trace)**.** *For every integer $k \geq 1$ and every $0 < r < R_0$, the following equality is exact:*

$$\boxed{\int_{-1}^{1} \frac{A(r,x)}{P(r,x)}\, U_{k-1}(x)\sqrt{1-x^2}\,\mathrm{d}x = \pi\, r^k q_k.} \tag{2.7}$$

No Riemann hypothesis, simplicity assumption, or positivity assumption on A is required. For arbitrary radii, the correct extension is the radial-split trace of chapter 3, not the expression $r^k q_k$ outside its convergence disk.

Proof. Fix $r < R_0$ and write $x = \cos\theta$, $0 < \theta < \pi$. By (2.6) and conjugation symmetry,

$$\begin{aligned} \log P(r, \cos\theta) &= 2\,\mathrm{Re}\log\Theta(re^{i\theta}) \\ &= 2\log\Theta(0) + 2\sum_{n=1}^{\infty} \frac{q_n}{n}\, r^n \cos(n\theta). \end{aligned} \tag{2.8}$$

The series and its differentiated series converge uniformly on the circle because $r < R_0$. Since $\partial_x = -\frac{1}{\sin\theta}\partial_\theta$, differentiation of (2.8) yields

$$\begin{aligned} \mathcal{L}(r,x) &= 2\sum_{n=1}^{\infty} q_n r^n \frac{\sin(n\theta)}{\sin\theta} \\ &= 2\sum_{n=1}^{\infty} q_n r^n U_{n-1}(x). \end{aligned} \tag{2.9}$$

Uniform convergence all the way to $x = \pm 1$ follows from $|U_{n-1}(x)| \leq n$ and absolute convergence of $\sum n|q_n|r^n$.

Multiply (2.9) by $U_{k-1}(x)\sqrt{1-x^2}$ and integrate. Equation (1.8) isolates the kth term:

$$\begin{aligned} \int_{-1}^{1} \mathcal{L}(r,x) U_{k-1}(x)\sqrt{1-x^2}\,\mathrm{d}x &= 2\sum_{n=1}^{\infty} q_n r^n \frac{\pi}{2}\,\mathbf{1}_{\{n=k\}} \\ &= \pi r^k q_k. \end{aligned}$$

Since $\mathcal{L} = A/P$ on the zero-free circle, this is (2.7). □

Reconstruction formula. The same proof gives a fully explicit local expression for the angular logarithmic derivative:

$$\frac{\partial_x P(r,x)}{P(r,x)} = 2\sum_{k=1}^{\infty} q_k r^k U_{k-1}(x), \qquad 0 < r < R_0. \tag{2.10}$$

The integral theorem is the exact coefficient-extraction rule for this series. Inverting (2.7) gives

$$q_k = \frac{1}{\pi r^k}\int_{-1}^{1} \mathcal{L}(r,x) U_{k-1}(x)\sqrt{1-x^2}\,\mathrm{d}x.$$

The right side is independent of which admissible radius r is chosen.

2.3 Why the factor r^k cannot be omitted

The kth monomial in $\log \Theta(z)$ is $q_k z^k/k$. Restricting it to $z = re^{i\theta}$ introduces r^k before either differentiation or integration. A version of (2.7) with πq_k is correct *only* if its symbol denotes a radius-dependent quantity $Q_k(r) = r^k q_k$, or if $r = 1$.

There are two immediate checks. For $k = 1$, $U_0 \equiv 1$ and

$$\int_{-1}^{1} \mathcal{L}(r, x)\sqrt{1 - x^2}\, \mathrm{d}x = \pi r q_1. \tag{2.11}$$

For $k = 2$, $U_1(x) = 2x$ and

$$\int_{-1}^{1} \mathcal{L}(r, x)\, 2x\sqrt{1 - x^2}\, \mathrm{d}x = \pi r^2 q_2. \tag{2.12}$$

2.4 The first two coefficients from derivatives of ξ

Because $\xi(\frac{1}{2} + w)$ is even,

$$\Theta(z) = \xi(\tfrac{1}{2}) + \frac{\xi''(\frac{1}{2})}{2!}\, z + \frac{\xi^{(4)}(\frac{1}{2})}{4!}\, z^2 + \cdots .$$

Comparing logarithms gives the exact identities

$$\begin{aligned} q_1 &= \frac{\xi''(\frac{1}{2})}{2\xi(\frac{1}{2})}, \\ q_2 &= \frac{\xi^{(4)}(\frac{1}{2})}{12\xi(\frac{1}{2})} - \left(\frac{\xi''(\frac{1}{2})}{2\xi(\frac{1}{2})}\right)^2 . \end{aligned} \tag{2.13}$$

In general, q_k is the coefficient of z^{k-1} in $\Theta'(z)/\Theta(z)$:

$$\frac{\Theta'(z)}{\Theta(z)} = \sum_{k \geq 1} q_k z^{k-1}, \quad q_k = \frac{1}{(k-1)!}\, \frac{\mathrm{d}^{k-1}}{\mathrm{d}z^{k-1}} \frac{\Theta'(z)}{\Theta(z)}\bigg|_{z=0}. \tag{2.14}$$

Thus the trace coefficients may be computed either from the nontrivial zero divisor or from derivatives of the *actual* completed zeta function at the symmetry center.

CHAPTER 3

The global radial-split Chebyshev trace

3.1 The radius of convergence is not a barrier to the trace

The expansion (2.10) stops at R_0 for a precise reason: the origin-centered logarithm $\log \Theta(z)$ encounters its first zero. It would be wrong to substitute $r > R_0$ into the same power series. Nonetheless, the *circlewise* Fourier transform of $\log|\Theta|^2$ makes sense at every positive radius, because logarithmic zero singularities are integrable.

For a transformed zero $\alpha \neq 0$, write

$$\alpha = R_\alpha e^{i\phi_\alpha}, \qquad R_\alpha = |\alpha| > 0, \quad \phi_\alpha \in (-\pi, \pi]. \tag{3.1}$$

All zero sums in this chapter count analytic multiplicities. The zero multiset is invariant under conjugation; hence the sine components of its complete Fourier sums cancel.

Definition 3.1 (Global radial-split trace)**.** For $k \geq 1$ and $r > 0$, define

$$\boxed{T_k(r) := -\sum_{\alpha \in \mathcal{Z}_\Theta} \left(\frac{\min\{r, R_\alpha\}}{\max\{r, R_\alpha\}}\right)^k \cos(k\phi_\alpha).} \tag{3.2}$$

Equivalently, with $u = \log r$ and $u_\alpha = \log R_\alpha$,

$$T_k(e^u) = -\sum_{\alpha \in \mathcal{Z}_\Theta} e^{-k|u-u_\alpha|} \cos(k\phi_\alpha). \tag{3.3}$$

This is a two-sided exponentially decaying kernel in *logarithmic radius*. It is defined on the actual, complete zero divisor, without assuming RH.

Lemma 3.2 (Absolute convergence and local agreement)**.** *For every $r > 0$ and $k \geq 1$,* (3.2) *converges absolutely. The function $r \mapsto T_k(r)$ is continuous on $(0, \infty)$. If $0 < r < R_0$, then*

$$\boxed{T_k(r) = r^k q_k.} \tag{3.4}$$

Proof. Only finitely many transformed zeros lie in $|\alpha| \leq r$. Their terms are finite. For $|\alpha| > r$, the absolute value of the summand is at most $(r/|\alpha|)^k \leq r/|\alpha|$; this is summable by (2.1). On every compact radial interval $r \in [a, b] \subset (0, \infty)$, remove the finitely many zeros with $|\alpha| \leq 2b$. The remaining tail is bounded uniformly by $b/|\alpha|$. Uniform convergence gives continuity. If $r < R_0$, all zeros are outside; conjugation symmetry makes $\sum_\alpha \alpha^{-k}$ real, giving $T_k(r) = -r^k \sum_\alpha |\alpha|^{-k} \cos(k\phi_\alpha) = r^k q_k$. □

3.2 A proof by splitting each canonical factor

For a single nonzero zero $\alpha = Re^{i\phi}$, the following identity is elementary whenever $r \neq R$:

$$\log\left|1-\frac{re^{i\theta}}{\alpha}\right|^2 = \begin{cases} -2\displaystyle\sum_{k\geq 1}\frac{(r/R)^k}{k}\cos k(\theta-\phi), & r<R, \\ 2\log(r/R) - 2\displaystyle\sum_{k\geq 1}\frac{(R/r)^k}{k}\cos k(\theta-\phi), & r>R. \end{cases} \tag{3.5}$$

The second line follows by extracting the factor r/R and using $|1-(R/r)e^{-i(\theta-\phi)}| = |1-(R/r)e^{i(\theta-\phi)}|$. At $r = R$, the Fourier series of $\log|1-e^{i(\theta-\phi)}|^2$ is understood in L^1; its cosine coefficients equal the limits of both lines. After summing over conjugate zeros, all $\sin(k\theta)$ contributions vanish.

Theorem 3.3 (Global logarithmic Chebyshev trace)**.** *For* every $r > 0$ *and integer* $k \geq 1$*, the integral is finite and*

$$\boxed{k\int_0^\pi \log P(r,\cos\theta)\cos(k\theta)\,\mathrm{d}\theta = \pi T_k(r).} \tag{3.6}$$

Here $\log P$ *is interpreted as an almost-everywhere function; its logarithmic singularities at the finitely many zeros on the circle are integrable. No assumption that the circle is zero-free is needed.*

Proof. Start with the genus-zero product (2.1) and separate the finitely many factors with $R_\alpha \leq 2r$ from the remaining factors. The far-zero logarithms converge absolutely and uniformly on $|z| = r$, because $\sum_\alpha |\alpha|^{-1} < \infty$. The finitely many factors inside the circle admit the second expansion in (3.5); those on the circle have integrable logarithmic singularities and the same Fourier coefficients by an L^1 limit.

For each α, the coefficient of $\cos(k\theta)$ is $-2\tau_\alpha(r)^k\cos(k\phi_\alpha)/k$ after pairing its complex conjugate, where $\tau_\alpha(r) = \min(r,R_\alpha)/\max(r,R_\alpha)$. The constant terms are orthogonal to $\cos(k\theta)$. Orthogonality $\int_0^\pi \cos^2(k\theta)\,\mathrm{d}\theta = \pi/2$ therefore gives the coefficient $\pi T_k(r)/k$, proving the identity. □

New master formula. The exact finite-radius moment $\pi r^k q_k$ is the *inner-radius branch* of a global object:

$$\boxed{\pi T_k(r) = -\pi\sum_{\alpha\in\mathcal{Z}_\Theta} e^{-k|\log(r/|\alpha|)|}\cos(k\arg\alpha).}$$

Crossing a zero radius changes the *radial representation*, not the validity of the logarithmic Fourier identity.

3.3 Recovery of the angular derivative at regular radii

A *regular radius* means $\Theta(z) \neq 0$ for every $|z| = r$. Such radii form the complement of the discrete set $\{|\alpha| : \alpha \in \mathcal{Z}_\Theta\}$.

Theorem 3.4 (Global derivative Chebyshev identity)**.** *At every regular radius* $r > 0$ *the Chebyshev series converges absolutely and uniformly for* $-1 \leq x \leq 1$*, and*

$$\frac{\partial_x P(r,x)}{P(r,x)} = 2\sum_{n=1}^\infty T_n(r)U_{n-1}(x), \tag{3.7}$$

$$\int_{-1}^1 \frac{\partial_x P(r,x)}{P(r,x)}U_{k-1}(x)\sqrt{1-x^2}\,\mathrm{d}x = \boxed{\pi T_k(r)}. \tag{3.8}$$

Proof. At a fixed regular radius, the quantities $\tau_\alpha(r)$ are all strictly smaller than 1 and have a common upper bound $\eta_r < 1$: only finitely many zeros lie in a bounded annulus, and $\tau_\alpha(r) \to 0$ along the unbounded zero set. Moreover $\sum_\alpha \tau_\alpha(r) < \infty$. Hence

$$\sum_{n\geq 1} n|T_n(r)| \leq \sum_{\alpha\in\mathcal{Z}_\Theta}\sum_{n\geq 1} n\tau_\alpha(r)^n \leq \frac{\sum_\alpha \tau_\alpha(r)}{(1-\eta_r)^2} < \infty.$$

The zero-free boundary allows differentiation of the full cosine expansion in (3.5). In the variable $x = \cos\theta$, its derivative is $2\sum_n T_n(r)U_{n-1}(x)$. The estimate above and $|U_{n-1}(x)| \leq n$ justify uniform convergence including the endpoints. Chebyshev orthogonality then yields (3.8). □

What happens on a zero circle? If $\Theta(re^{i\theta_0}) = 0$ with $0 < \theta_0 < \pi$, then P_x/P has a first-order pole at $x_0 = \cos\theta_0$; its unqualified integral need not exist. The logarithmic identity (3.6) remains exact. At the endpoint $\theta = \pi$ (a critical-line zero), a weighted improper integral can still exist, but the uniform differentiated series is not claimed there.

3.4 Jensen's formula and the radial zero count

The zeroth Fourier coefficient is as informative as the higher ones. Define

$$J(r) := \frac{1}{2\pi}\int_0^{2\pi} \log|\Theta(re^{i\theta})|\,\mathrm{d}\theta.$$

The same factor splitting gives Jensen's exact identity

$$\boxed{J(r) = \log|\Theta(0)| + \sum_{|\alpha|<r} \log\frac{r}{|\alpha|}.} \tag{3.9}$$

The summand vanishes at equality, so (3.9) is valid also when the circle contains zeros. At regular radii,

$$\boxed{rJ'(r) = N_\Theta(r), \quad N_\Theta(r) := \#\{\alpha \in \mathcal{Z}_\Theta : |\alpha| < r\},} \tag{3.10}$$

with multiplicities. Thus one circle profile carries *two* exact spectral readouts: its mean logarithm counts zeros radially, while its nonzero Chebyshev moments encode their angles through $T_k(r)$.

3.5 Radial continuity, kinks, and a one-factor check

By theorem 3.2, $T_k(r)$ is continuous even when a circle meets zeros. At a zero radius R it has a predictable derivative jump:

$$\boxed{T_k'(R^+) - T_k'(R^-) = \frac{2k}{R}\sum_{|\alpha|=R}\cos(k\phi_\alpha).} \tag{3.11}$$

Indeed, the α summand in (3.2) is $-\cos(k\phi_\alpha)(r/R)^k$ below R and $-\cos(k\phi_\alpha)(R/r)^k$ above R. Every other zero contributes a smooth term near R; the displayed jump follows by differentiating.

Example 3.5 (The critical-line factor across its zero radius)**.** Let $G(z) = 1 + z/a$ with $a > 0$. It has one transformed zero $\alpha = -a$, and

$$T_k^G(r) = (-1)^{k+1}\left(\frac{\min\{r,a\}}{\max\{r,a\}}\right)^k. \tag{3.12}$$

For $r < a$, this is the original coefficient $q_k r^k$. At $r > a$, it becomes $(-1)^{k+1}(a/r)^k$. At $r = a$, the Fourier coefficient is still well-defined though $P_G(a, -1) = 0$. The origin-centered Taylor expansion cannot yield the exterior expression, but the global radial-split theorem does.

Interpretation. The identity on $r < R_0$ is a Taylor-coefficient theorem. The identity on arbitrary circles is a *radial Fourier theorem*, not analytic continuation of the same Taylor series. Neither theorem by itself forces $P_x \geq 0$; that sign is the separate QQRH problem.

CHAPTER 4

Why the radius 255/7 is a genuine anchor

4.1 A low-height input and zero-free disk

The number 255/7 is *not* the analytic radius of convergence of the Chebyshev series. It is a convenient, rigorously derived radius for a stronger *angular monotonicity* result. Its source is the following low-height input.

Proposition 4.1 (Low-height zero gap)**.** *Every zero $\rho = \beta + i\gamma$ of ξ satisfies*

$$0 < \beta < 1, \qquad |\gamma| > \frac{13}{2}. \tag{4.1}$$

The critical-strip localization is classical. A proof of the second inequality is reported in the supplied QQRH Full-x Anchor 2 archive (`AnchorHeight.lean`). Its analytic intermediate bound is

$$\left|\mathrm{completedRiemannZeta}_0(s)\right| \le \frac{9}{385}, \qquad 0 \le \operatorname{Re} s \le 1, \tag{4.2}$$

where this *pole-subtracted* completion is normalized so that

$$\xi(s) = \tfrac{1}{2}\big(1 + s(s-1)\,\mathrm{completedRiemannZeta}_0(s)\big). \tag{4.3}$$

The theta–Mellin kernel estimate leading to (4.2) is summarized in Appendix A; it is a separately identified analytic input rather than a result silently assumed from RH.

Deduction of the zero gap from (4.2). For $0 \le \beta \le 1$ and $|\gamma| \le 13/2$ one has

$$|s(s-1)| \le \gamma^2 + \tfrac{1}{2} \le \frac{171}{4}.$$

Therefore (4.2) gives

$$|s(s-1)\,\mathrm{completedRiemannZeta}_0(s)| \le \frac{171}{4}\frac{9}{385} = \frac{1539}{1540} < 1.$$

Equation (4.3) then makes $\xi(s) = 0$ impossible. □

Corollary 4.2 (A safe disk for the Chebyshev identity)**.** *For every $\alpha \in \mathcal{Z}_\Theta$,*

$$|\alpha| = |\rho - \tfrac{1}{2}|^2 = (\beta - \tfrac{1}{2})^2 + \gamma^2 > \frac{169}{4}. \tag{4.4}$$

Consequently, the exact Chebyshev identity holds for all $k \ge 1$ and

$$\boxed{0 < r \le \frac{255}{7} < \frac{169}{4} \le R_0.}$$

This already proves the finite-radius *integral* identity. The next sections explain why 255/7, which is below the simple zero-free radius 169/4, also carries angular derivative positivity.

4.2 One off-axis conjugate pair: exact factor geometry

Suppose a nonreal transformed zero is

$$\alpha = (\delta + i\gamma)^2, \quad g := |\gamma| > 0, \quad h := |\delta| \le \tfrac{1}{2}, \quad D := g^2 + h^2 = |\alpha|.$$

By reality, its conjugate $\overline{\alpha}$ is also a zero. Introduce

$$q = \frac{r}{D}, \qquad c = \frac{\operatorname{Re}\alpha}{|\alpha|} = \frac{h^2 - g^2}{g^2 + h^2} \in (-1, 0). \tag{4.5}$$

On the circle $z = z_r(x)$, the paired factor contributes

$$\begin{aligned} B_{q,c}(x) &= \left|\left(1 - \frac{z}{\alpha}\right)\left(1 - \frac{z}{\overline{\alpha}}\right)\right|^2 \\ &= (1 + q^2)^2 - 4qc(1 + q^2)x + 4q^2\big(x^2 + c^2 - 1\big). \end{aligned} \tag{4.6}$$

No approximation has been made. Differentiating gives

$$B'_{q,c}(x) = 4q\big(2qx - c(1 + q^2)\big). \tag{4.7}$$

The derivative is smallest at $x = -1$. Therefore $B_{q,c}$ is nondecreasing on $[-1, 1]$ precisely when

$$2q + c(1 + q^2) \le 0. \tag{4.8}$$

Because $-c = (g^2 - h^2)/(g^2 + h^2)$, the first nonnegative root of the associated quadratic is

$$q_- = \frac{g - h}{g + h}.$$

For $0 \le q \le q_-$, inequality (4.8) holds; equivalently,

$$\boxed{0 \le r \le H(g, h) := (g^2 + h^2)\frac{g - h}{g + h}.} \tag{4.9}$$

This is the *exact inner-radius factor criterion* for an off-axis conjugate pair.

4.3 The full forbidden radial band of a nonreal pair

The inner threshold $q_- = \frac{g-h}{g+h}$ is only one branch of the exact condition

$$B'_{q,c}(x) \ge 0 \quad \text{for all } x \in [-1, 1].$$

For $h > 0$, the quadratic inequality (4.8) has a second root,

$$q_+ = \frac{g + h}{g - h} = \frac{1}{q_-} > 1.$$

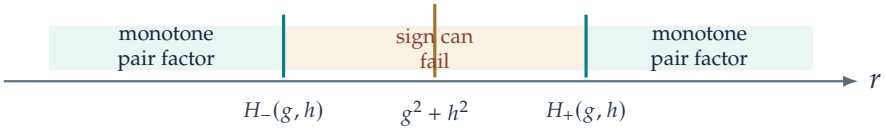
Since its leading coefficient is positive, the complete pair criterion is

$$\boxed{B'_{q,c}(x) \ge 0 \text{ on } [-1, 1] \iff 0 \le q \le q_- \text{ or } q \ge q_+.} \tag{4.10}$$

Thus a single off-axis pair loses full-angle monotonicity only in the radial annulus

$$\boxed{(g^2 + h^2)\frac{g - h}{g + h} < r < (g^2 + h^2)\frac{g + h}{g - h}.} \tag{4.11}$$

The zero radius $r = |\alpha| = g^2 + h^2$ lies strictly inside this annulus. The failure of the *pair's* derivative sign in that annulus does not by itself prove failure for the full infinite product; a genuine nonreal zero gives the stronger local obstruction proved later in theorem 5.5.



4.4 Minimizing over the critical strip

It remains to find a uniform lower bound for $H(g,h)$ when

$$g \geq \frac{13}{2}, \quad 0 \leq h \leq \frac{1}{2}.$$

Elementary differentiation gives

$$\begin{aligned} \partial_g H &= \frac{2(g^3+g^2h-gh^2+h^3)}{(g+h)^2} > 0, \\ \partial_h H &= -\frac{2\big((g-h)g^2+(g+h)h^2\big)}{(g+h)^2} \leq 0. \end{aligned} \tag{4.12}$$

In this domain $g > h \geq 0$, so the minimum is at $g = 13/2$, $h = 1/2$:

$$\boxed{H(g,h) \geq H(\tfrac{13}{2},\tfrac{1}{2}) = \left(\frac{169}{4}+\frac{1}{4}\right)\frac{6}{7} = \frac{255}{7}.} \tag{4.13}$$

In particular every off-axis zero pair contributes a nondecreasing positive factor for $0 \leq r \leq 255/7$. A transformed zero on the negative real axis, $\alpha = -g^2$, instead contributes the single factor

$$\left|1+\frac{z_r(x)}{g^2}\right|^2 = 1+2\frac{r}{g^2}x+\frac{r^2}{g^4}, \tag{4.14}$$

whose x derivative is $2r/g^2 \geq 0$. Both cases therefore have the desired sign.

Theorem 4.3 (Full-angle finite-radius monotonicity)**.** *Subject to the low-height input* (4.1) *(which is supplied by the theta–Mellin estimate described above), for every* $0 < r \leq 255/7$,

$$\boxed{P(r,x) > 0 \quad (-1 \leq x \leq 1), \qquad \partial_x P(r,x) \geq 0 \quad (-1 < x < 1).} \tag{4.15}$$

In particular, $P(r,\cdot)$ *is nondecreasing on the entire closed interval* $[-1,1]$.

Proof. Group the genus-zero product (2.1) into real negative factors (4.14) and nonreal conjugate pairs (4.6). Every factor is nonnegative and nondecreasing in x on the stated radial interval, by (4.8)–(4.13). Every *finite* product is therefore nondecreasing. Local uniform convergence of the canonical product passes this monotonicity to $P(r,x) = |\Theta(z_r(x))|^2$. The zero gap gives $|z_r(x)| = r \leq 255/7 < 169/4 < |\alpha|$ for every $\alpha \in \mathcal{Z}_\Theta$, hence $P(r,x) > 0$. Finally the interior derivative of a differentiable nondecreasing function is nonnegative. □

Why this is stronger than merely having no zeros in the disk. From $|\alpha| > 169/4$ alone, one gets $P > 0$ for $r \leq 255/7$. The additional nontrivial statement $\partial_x P \geq 0$ uses the conjugate-factor vertex inequality (4.8). The special number $255/7$ comes from the uniform minimum (4.13), not from a first-zero table.

4.5 A height-to-radius amplification theorem

The special choice 13/2 can be replaced by *any* certified height up to which off-line zeros have been excluded. This is the exact link between rigorous zero-counting experiments and the geometric anchor.

Definition 4.4 (Exhaustive height certificate)**.** For $T > \frac{1}{2}$, write $\mathsf{EH}(T)$ for the assertion

$$\forall \rho = \beta + i\gamma \quad \big(\xi(\rho) = 0 \ \wedge \ |\gamma| \le T\big) \Longrightarrow \beta = \tfrac{1}{2}. \tag{4.16}$$

This is an *all-zeros* statement below height T, not merely a list of zeros found on the critical line. To establish it computationally, a rigorous zero count must match the number of critical-line zeros (including multiplicity).

Theorem 4.5 (Height-to-radius angular anchor)**.** *Let $T > \frac{1}{2}$ and assume $\mathsf{EH}(T)$. Set*

$$\boxed{R(T) := \left(T^2 + \frac{1}{4}\right)\frac{T - \frac{1}{2}}{T + \frac{1}{2}}.} \tag{4.17}$$

Then, for every $0 < r \le R(T)$,

$$P(r, x) > 0 \ \ (-1 < x \le 1), \qquad \partial_x P(r, x) \ge 0 \ \ (-1 < x < 1). \tag{4.18}$$

In addition $P(r, \cdot)$ is nondecreasing on the full closed interval $[-1, 1]$. It may vanish at $x = -1$ if $r = \gamma^2$ for an already certified critical-line zero.

Proof. A transformed zero arising from an on-line zero is $\alpha = -\gamma^2$ and contributes the nondecreasing factor (4.14) for every radius. An off-line pair has $h = |\beta - \frac{1}{2}| \in (0, \frac{1}{2})$ and $g = |\gamma| > T$ by $\mathsf{EH}(T)$. The monotonicity of $H(g, h)$ in (4.12) yields $H(g, h) \ge H(T, \frac{1}{2}) = R(T)$. By (4.8) every off-line paired factor is nondecreasing for $r \le R(T)$ and has no zero on the circle (since $r \le H(g, h) < g^2 + h^2$). Partial products of nonnegative nondecreasing factors are nondecreasing; pass to the locally uniform genus-zero product. On-line factors vanish only at $x = -1$ when $r = \gamma^2$, so the full profile is strictly positive for $-1 < x \le 1$. Interior differentiability supplies $P_x \ge 0$. □

Corollary 4.6 (The previous anchor as the first instance)**.** *The zero-height statement $|\gamma| > 13/2$ implies $\mathsf{EH}(13/2)$ vacuously. Substitution into* (4.17) *gives*

$$R(13/2) = \left(\frac{169}{4} + \frac{1}{4}\right)\frac{6}{7} = \frac{255}{7}.$$

Under the stronger height-14 certificate $\mathsf{EH}(14)$, one obtains

$$R(14) = \left(196 + \frac{1}{4}\right)\frac{27}{29} = \frac{21195}{116} \approx 182.7155.$$

The second number is an illustration of the theorem*: it is justified whenever the stated exhaustive height certificate is available. Neither value is a proof at arbitrarily large radius.*

Remark 4.7 (Scaling and the limiting obstruction)*.* A direct expansion gives

$$R(T) = T^2 - T + \frac{3}{4} + O(T^{-1}) \qquad (T \to \infty). \tag{4.19}$$

Thus an exhaustive zero classification to height T yields angular monotonicity out to a squared radius of order T^2. This does *not* let one conclude RH from any fixed finite T. If $\mathsf{EH}(T)$ were established for an unbounded sequence $T \to \infty$, it would already imply RH; the radius theorem then supplies an equivalent geometric exhaustion.

CHAPTER 5

The QQRH angular-exclusion bridge

5.1 From the critical strip to an angular cutoff

Write a nontrivial zero as

$$\rho = \beta + i\gamma, \quad \delta = \beta - \tfrac{1}{2}, \qquad z = (\rho - \tfrac{1}{2})^2 = (\delta + i\gamma)^2.$$

If $r = |z| = \delta^2 + \gamma^2$, then the angular cosine is

$$x = \frac{\operatorname{Re} z}{|z|} = \frac{\delta^2 - \gamma^2}{\delta^2 + \gamma^2}. \tag{5.1}$$

The mapping has a useful exact inverse at any transformed zero:

$$\boxed{|\delta| = \sqrt{\frac{r(1+x)}{2}}, \qquad |\gamma| = \sqrt{\frac{r(1-x)}{2}}, \qquad \frac{|\gamma|}{|\delta|} = \sqrt{\frac{1-x}{1+x}}.} \tag{5.2}$$

Consequently, a critical-line zero sits at $x = -1$, while every off-line zero in the critical strip corresponds to an interior angle $x > -1$.

For any slope $m > 0$, set

$$a_m := \frac{1 - m^2}{1 + m^2} = -1 + \frac{2}{1 + m^2}. \tag{5.3}$$

Cross-multiplying positive denominators shows

$$\boxed{x > a_m \quad \Longleftrightarrow \quad |\gamma| < m|\delta|.} \tag{5.4}$$

Thus a sector $x > a_m$ in the squared-coordinate plane corresponds *exactly* to the wedge of slope m about the horizontal centerline in the (δ, γ) plane.

Definition 5.1 (QQRH at slope m)**.** For $m > 0$, QQRH(m) asserts that every nontrivial zero $\rho = \beta + i\gamma$ satisfies

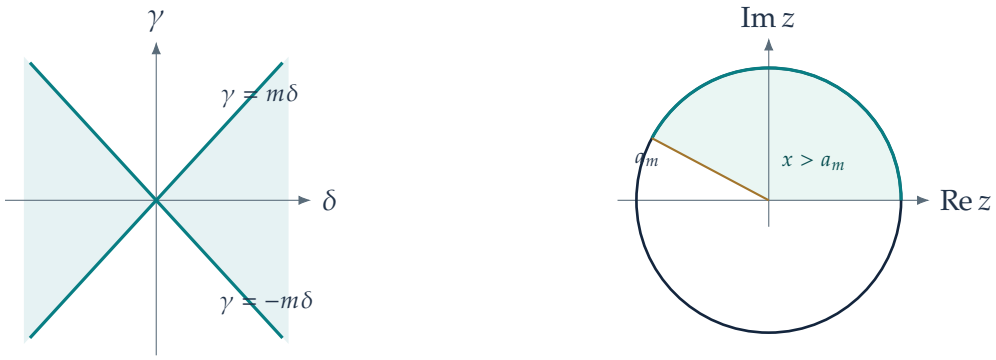
$$|\gamma| \geq m\,|\beta - \tfrac{1}{2}|. \tag{5.5}$$

Equivalently, the open wedge $|\gamma| < m|\beta - \frac{1}{2}|$ contains no nontrivial zeros. This is the formulation used in the supplied QQRH Lean package.

5.2 Why derivative positivity excludes a zero

Lemma 5.2 (The monotone-arc exclusion principle)**.** *Let G be a nonzero entire function and define $P_G(r, x) = |G(z_r(x))|^2$. Fix $r > 0$ and $a \in [-1, 1)$. If $P_G(r, \cdot)$ is nondecreasing on $(a, 1)$, then*

$$G(z_r(x)) \neq 0 \qquad \textit{for every } a < x \leq 1.$$



Critical-strip wedge **Squared-coordinate sector**

Figure 5.1: The square map $z = (s - \frac{1}{2})^2$ converts the zero-exclusion wedge into a sector. Angular boundaries are schematic, not to scale.

Proof. If $G(z_r(x_0)) = 0$ at some $x_0 \in (a, 1]$, then $P_G(r, x_0) = 0$. For any $a < x < x_0$, monotonicity and $P_G \geq 0$ force

$$0 \leq P_G(r, x) \leq P_G(r, x_0) = 0.$$

Hence G vanishes on a nondegenerate arc of the circle $|z| = r$. That arc has a limit point in $\mathbb{C}$, so the identity theorem forces $G \equiv 0$, a contradiction. If $x_0 = 1$, the same argument uses the continuous endpoint value; monotonicity extends to that endpoint by continuity. □

Theorem 5.3 (Global angular positivity implies QQRH)**.** *Let $m > 0$ and a_m be as in* (5.3)*. If*

$$\boxed{\forall r > 0\ \forall x \in (a_m, 1): \quad \partial_x P(r, x) \geq 0,} \tag{5.6}$$

then QQRH(m) *holds.*

Proof. For each $r > 0$, nonnegative derivative on $(a_m, 1)$ implies that $P(r, \cdot)$ is nondecreasing there. Lemma 5.2 says the corresponding sector contains no zero of Θ. If $\xi(\rho) = 0$ and $|\gamma| < m|\delta|$, (1.4) and (5.4) put the corresponding zero of Θ in that sector, a contradiction. □

Corollary 5.4 (Global spectral criterion, wherever the series applies)**.** *For $0 < r < R_0$, the sign statement $\partial_x P(r, x) \geq 0$ is equivalent to*

$$\sum_{k=1}^{\infty} q_k r^k U_{k-1}(x) \geq 0. \tag{5.7}$$

To obtain QQRH(m) *using Theorem 5.3, this inequality (or a replacement valid beyond the disk) must be certified for* every *$r > 0$, not only $r < R_0$.*

5.3 A nonreal zero forces a local sign reversal

The monotone-arc principle has a sharp converse at any nonreal zero, with no simplicity assumption. This rules out a common misinterpretation: one does *not* need to force the zero to extend along an analytic zero arc. A finite-order point zero already produces the requisite sign failure.

Theorem 5.5 (Local angular obstruction at arbitrary multiplicity)**.** *Suppose $\Theta(\alpha) = 0$ with $\Im\alpha > 0$ and analytic multiplicity $\nu \geq 1$. Set $r_0 = |\alpha|$ and $x_0 = \Re\alpha/|\alpha| \in (-1, 1)$. Then for a constant $C > 0$,*

$$P(r_0, x) = C(x - x_0)^{2\nu}\big(1 + O(x - x_0)\big), \tag{5.8}$$

$$\partial_x P(r_0, x) = 2\nu C(x - x_0)^{2\nu-1}\big(1 + O(x - x_0)\big). \tag{5.9}$$

In particular, for some $\varepsilon > 0$,

$$\boxed{P_x(r_0, x) < 0 \quad (x_0 - \varepsilon < x < x_0), \qquad P_x(r_0, x) > 0 \quad (x_0 < x < x_0 + \varepsilon).} \tag{5.10}$$

Moreover the negative sign persists in a two-dimensional open set of nearby regular radii and angles.

Proof. The nonzero holomorphic germ has a local factorization $\Theta(z) = (z - \alpha)^\nu H(z)$ with $H(\alpha) \neq 0$. On the upper circle, $z'_{r_0}(x_0) = r_0(1 - ix_0/\sqrt{1 - x_0^2}) \neq 0$, so $z_{r_0}(x) - \alpha = (x - x_0)z'_{r_0}(x_0) + O((x - x_0)^2)$. Taking the squared modulus gives (5.8) with $C = |H(\alpha)|^2 |z'_{r_0}(x_0)|^{2\nu} > 0$. Differentiating the real-analytic expansion yields (5.9), and the odd exponent $2\nu - 1$ gives opposite signs. Choose any point $x \in (x_0 - \varepsilon, x_0)$, where the negative value is strict; continuity of P_x in r, x preserves it for nearby r. Since zero moduli form a discrete set, these nearby radii can be chosen regular. □

Zero geometry is enough. At an off-line zero, $P = P_x = 0$ and the first nonzero angular Taylor term has *even* order 2ν. This is a point-contact phenomenon; it creates a negative derivative immediately to the left without requiring any neighboring zeros or a zero arc.

5.4 An exact positivity characterization of RH

Theorem 5.6 (RH if and only if global full-angle positivity)**.** *For the* actual *completed zeta function the following statements are equivalent:*

(i) RH is true;

(ii) $\partial_x P(r, x) \geq 0$ for every $r > 0$ and $-1 < x < 1$;

(iii) $\partial_x P(r, x) > 0$ for every $r > 0$ and $-1 < x < 1$.

Proof. Assume RH. Every transformed zero is $-\gamma^2$, with $\gamma > 0$. The genus-zero product reads

$$\Theta(z) = \Theta(0) \prod_{\gamma > 0} \left(1 + \frac{z}{\gamma^2}\right)^{m_\gamma}.$$

For any $r > 0$ and $-1 < x < 1$, all circle factors are strictly positive. Differentiating their locally uniformly convergent logarithmic product gives

$$\boxed{\frac{P_x(r, x)}{P(r, x)} = \sum_{\gamma > 0} m_\gamma \frac{2r/\gamma^2}{1 + 2(r/\gamma^2)x + (r/\gamma^2)^2} > 0.} \tag{5.11}$$

Absolute convergence follows from $\sum_\gamma m_\gamma/\gamma^2 < \infty$; each summand is positive. This establishes (iii), and (iii) implies (ii).

Conversely assume (ii). If RH were false, some zero would have $\beta \neq 1/2$, giving a nonreal transformed zero. Conjugation supplies one in the upper half-plane. By theorem 5.5 its circle profile has $P_x < 0$ at an interior angle, contradicting (ii). Thus (ii) implies RH. □

The exact analytic frontier.

$$\boxed{\text{RH} \iff \forall r > 0 \; \forall x \in (-1, 1) : \; P_x(r, x) \geq 0.}$$

This is a *new formulation of the objective*, not its solution. The finite anchor supplies precisely the part $r \leq 255/7$ of this quantified requirement.

5.5 From slope exhaustion to RH

Proposition 5.7 (The precise closure condition)**.** *The following are equivalent:*

(i) *the Riemann hypothesis;*

(ii) QQRH(m) *for every* $m > 0$*;*

(iii) QQRH(m_j) *for some unbounded sequence* $m_j \to \infty$*.*

Proof. RH makes $|\beta - \frac{1}{2}| = 0$, so every slope condition holds. Conversely, if there were a zero with $\delta = \beta - \frac{1}{2} \neq 0$, its ratio $|\gamma|/|\delta|$ would be finite. Choose m_j larger than that ratio; QQRH(m_j) would fail. □

The angular cutoffs exhaust the full open interval because $a_m \downarrow -1$ as $m \to \infty$. Notice the exact residual angular gap

$$1 + a_m = \frac{2}{1 + m^2}. \tag{5.12}$$

For example

$$a_2 = -\frac{3}{5}, \qquad a_{11} = -\frac{60}{61}, \qquad 1 + a_{11} = \frac{1}{61}.$$

This 1/61 describes an angular *parameter gap*, not a measured percentage of a proof of RH.

CHAPTER 6

Coefficient certificates and the remaining frontier

6.1 An explicitly bounded spectral tail

It is tempting to replace the infinitely many sign constraints in (5.7) by a finite computation. The exact identity allows a mathematically honest finite-certificate statement, provided the tail is bounded.

Fix $N \geq 1$ and $0 < r < R_0$. Define the finite Chebyshev sum

$$S_N(r,x) = \sum_{k=1}^{N} q_k r^k U_{k-1}(x).$$

Using $|U_{k-1}(x)| \leq k$ on $[-1,1]$ and (2.5),

$$\left| \sum_{k>N} q_k r^k U_{k-1}(x) \right| \leq \sum_{\alpha \in \mathcal{Z}_\Theta} \sum_{k>N} k \left(\frac{r}{|\alpha|} \right)^k. \tag{6.1}$$

For $0 \leq t < 1$,

$$\sum_{k=N+1}^{\infty} k t^k = t^{N+1} \frac{(N+1) - Nt}{(1-t)^2}. \tag{6.2}$$

Therefore the *uniform, explicit* spectral tail is bounded by

$$\boxed{C_N(r) := \sum_{\alpha \in \mathcal{Z}_\Theta} \left(\frac{r}{|\alpha|} \right)^{N+1} \frac{(N+1) - N(r/|\alpha|)}{(1 - r/|\alpha|)^2}.} \tag{6.3}$$

It is finite for $r < R_0$, using $\sum |\alpha|^{-1} < \infty$.

Proposition 6.1 (Finite-degree sufficient certificate)**.** *Let $I \subseteq [-1,1]$ be an interval, $0 < r < R_0$ and $N \geq 1$. If*

$$S_N(r,x) \geq C_N(r) \qquad \text{for every } x \in I, \tag{6.4}$$

then $\partial_x P(r,x) \geq 0$ for every interior point of I.

Proof. The tail bound implies $\sum_{k \geq 1} q_k r^k U_{k-1}(x) \geq S_N(r,x) - C_N(r) \geq 0$. Multiply by $2P(r,x) > 0$ and use (2.10). □

Certificate versus criterion. Equation (6.4) is *sufficient*, not automatic. The sum $C_N(r)$ must itself be bounded rigorously, and the entire method is currently restricted to $r < R_0$. No finite number of moments with an uncontrolled tail proves angular positivity for all radii.

6.2 Spectral certificates at arbitrary regular radii

The global trace from chapter 3 allows finite-degree error control *at any regular radius*, not only below R_0. Define

$$S_N^{\mathrm{glob}}(r,x) := \sum_{k=1}^{N} T_k(r)U_{k-1}(x), \qquad \tau_\alpha(r) := \frac{\min(r,|\alpha|)}{\max(r,|\alpha|)}.$$

At a regular radius, $0 \le \tau_\alpha(r) < 1$ for every zero. Applying (6.2) yields

$$\boxed{C_N^{\mathrm{glob}}(r) := \sum_{\alpha\in\mathcal{Z}_\Theta} \tau_\alpha(r)^{N+1}\frac{(N+1)-N\tau_\alpha(r)}{(1-\tau_\alpha(r))^2} < \infty.} \tag{6.5}$$

Proposition 6.2 (All-radius finite-degree sign certificate)**.** *Fix a regular radius $r > 0$, an interval $I \subseteq [-1,1]$, and an integer $N \ge 1$. If*

$$S_N^{\mathrm{glob}}(r,x) \ge C_N^{\mathrm{glob}}(r) \qquad \text{for all } x \in I,$$

then $P_x(r,x) \ge 0$ for all interior $x \in I$.

Proof. The estimate $|U_{k-1}| \le k$ and the absolute bound $|T_k(r)| \le \sum_\alpha \tau_\alpha(r)^k$ show that the truncated tail is at most $C_N^{\mathrm{glob}}(r)$ in absolute value. Thus the full sum in (3.7) is nonnegative. Multiplication by $2P(r,x) > 0$ proves the claim. □

Remark 6.3 (Recovery of the finite-radius bound). If $r < R_0$ then $\tau_\alpha(r) = r/|\alpha|$, $T_k(r) = r^k q_k$, and (6.5) reduces to (6.3). The old local certificate is therefore an exact special case.

> **An exact formula is not an effective certificate by itself.** Computing or bounding $T_k(r)$ and $C_N^{\mathrm{glob}}(r)$ for the *complete* actual zero divisor is still required. Unknown zeros cannot be omitted. A rigorous numerical certificate must bound the unknown tail independently, especially near the radii of potential zeros where $\tau_\alpha(r) \approx 1$.

6.3 A global spectral formulation of the RH frontier

At every regular radius, (3.7) gives the exact equivalence

$$P_x(r,x) \ge 0 \quad\Longleftrightarrow\quad \sum_{k\ge1} T_k(r)U_{k-1}(x) \ge 0.$$

In fact theorem 5.5 shows that one can omit zero radii from the test, without losing the RH conclusion:

Corollary 6.4 (Global spectral positivity criterion)**.** *RH is equivalent to*

$$\boxed{\forall r > 0\ \big(\Theta \ne 0 \text{ on } |z| = r\big)\ \forall x \in (-1,1): \quad \sum_{k=1}^{\infty} T_k(r)U_{k-1}(x) \ge 0.} \tag{6.6}$$

Proof. Under RH, (5.11) gives strict positivity, and the global reconstruction applies at all regular radii. Conversely the condition gives $P_x \ge 0$ at all regular radii. If an off-line zero existed, theorem 5.5 would produce a nearby *regular* radius and angle with $P_x < 0$, contradiction. □

6.4 Two elementary models

Example 6.5 (One critical-line factor)**.** Let $G(z) = 1 + z/a$ with $a > 0$. This models one transformed critical-line zero $-a$. Then

$$P_G(r, x) = 1 + 2(r/a)x + (r/a)^2,$$
$$\partial_x \log P_G(r, x) = \frac{2(r/a)}{1 + 2(r/a)x + (r/a)^2} > 0,$$

whenever the denominator is nonzero. Its trace coefficients are

$$q_k = (-1)^{k+1} a^{-k}.$$

The coefficients alternate in sign even though the full angular derivative is positive. Consequently, requiring *each* $q_k \geq 0$ would be an incorrect positivity criterion.

Example 6.6 (One off-axis conjugate pair)**.** Let $G(z) = (1 - z/\alpha)(1 - z/\overline{\alpha})$ with nonreal α. Then

$$q_k = -\alpha^{-k} - \overline{\alpha}^{-k} = -2|\alpha|^{-k} \cos(k \arg \alpha).$$

The profile is exactly the quadratic (4.6). It is monotone across the full angular interval at the inner radii characterized by (4.9), while monotonicity of that individual factor can fail near $r = |\alpha|$. This is the geometry that makes *radial extension* difficult when off-axis zeros have not been ruled out.

6.5 What the two anchors cover — and what they do not

Suppose we combine the finite full-angle anchor in Theorem 4.3 with a separately established all-radius sector result $\partial_x P \geq 0$ on $x > a_m$. Their *combined sign domain* contains

$$\left\{(r, x) : 0 < r \leq \tfrac{255}{7},\ -1 < x < 1\right\} \cup \left\{(r, x) : r > 0,\ a_m < x < 1\right\}. \tag{6.7}$$

What remains uncertified by these two pieces is

$$\boxed{r > \frac{255}{7}, \qquad -1 < x \leq a_m.} \tag{6.8}$$

For the reported sector $m = 11$, its angle extends nearly to -1 but still leaves a thin open sector adjacent to the negative real axis at large radii.

6.6 A two-axis exhaustion blueprint

There are two logically distinct ways to close the uncovered region.

1. **Radial exhaustion:** prove $P_x \geq 0$ for $r \leq R_j$ and every $-1 < x < 1$, where $R_j \to \infty$. The finite 255/7 theorem is an initial member of this pattern. Its height-to-radius generalization proves such bounds conditional on exhaustive zero classification through heights T_j, but proving such classification for unbounded T_j is already equivalent to RH.
2. **Angular exhaustion:** prove $P_x \geq 0$ for every $r > 0$ and $a_{m_j} < x < 1$, where $m_j \to \infty$. Each such theorem yields QQRH(m_j), so the wedge exclusions exhaust the off-line possibilities.

A mixed approach can certify a growing union of rectangular subdomains $\{r \leq R_j,\ x > a_j\}$, provided the union covers *every* $r > 0$ and every $-1 < x < 1$. Covering a large numerical fraction or approaching the boundary at fixed finite radius is insufficient.

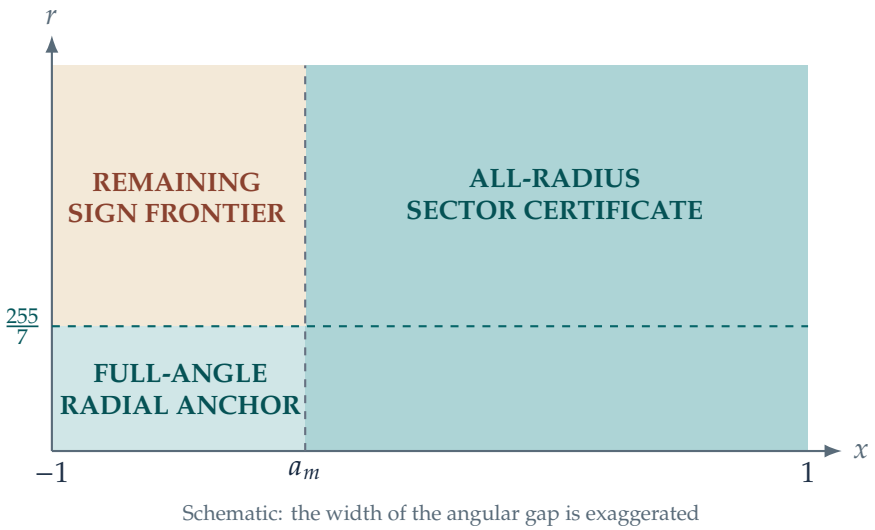


Figure 6.1: The radial and angular sign domains intersect. Their union does not cover every $r > 0$ and $-1 < x < 1$. The missing region is exactly (6.8).

6.7 A useful comparison: slopes below 13

The zero-height gap (4.1) has an immediate consequence that should be kept distinct from the stronger *sign* assertion.

Proposition 6.7 (A direct QQRH consequence of the height gap)**.** *For every* $0 < m \le 13$, QQRH(m) *follows directly from* (4.1) *and* $0 < \beta < 1$, *without an angular derivative argument.*

Proof. Since $|\beta - \frac{1}{2}| < \frac{1}{2}$ and $|\gamma| > \frac{13}{2}$, we have

$$|\gamma| > 13|\beta - \tfrac{1}{2}| \ge m|\beta - \tfrac{1}{2}|.$$

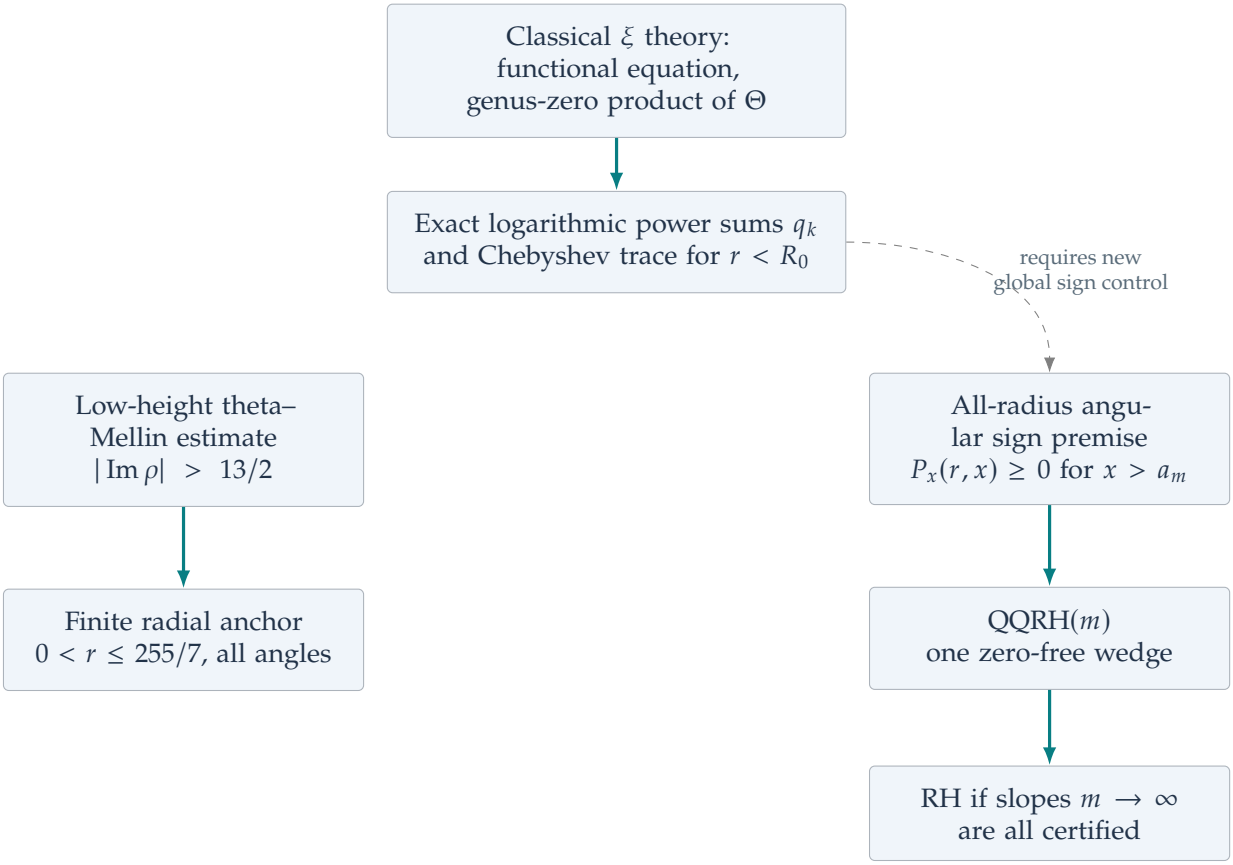
Thus the forbidden wedge condition is impossible. □

The reported $m = 11$ angular sign theorem is a substantive statement about P_x. But the *zero-exclusion conclusion* QQRH(11) is already a consequence of the weaker low-height gap. Progress toward RH requires extending the exclusion to an unbounded collection of slopes, or providing an independent global zero-location principle; merely reaching one more small finite slope does not close RH.

CHAPTER 7

Logical summary and research checklist

7.1 A precise implication diagram



7.2 What is proved, and what is still required

The second edition adds the global zero-aware identity (3.6), its derivative version (3.8), Jensen's radial counting identity (3.9), and the full-angle equivalence theorem 5.6. These are mathematical derivations in the present text, not additional claims of Lean formalization.

Statement	Precise status in this chapter
(2.7): local trace	Fully derived for every $0 < r < R_0$ from the genus-zero product.
Global radial-split trace	Fully derived for all $r > 0$ in log-Fourier form; derivative form on regular circles.
RH–monotonicity equivalence	Both directions are proved in theorem 5.6; this is a criterion, not a verification of its universal sign premise.
Height-to-radius theorem	Fully derived conditional on an exhaustive zero-height certificate $\mathsf{EH}(T)$.
Finite 255/7 domain	Follows from the low-height input $\lvert\gamma\rvert > 13/2$, via (4.4).
Full-angle sign anchor	Factor geometry and infinite-product deduction are derived in theorem 4.3; the low-height analytic estimate is sourced separately.
QQRH implication	Proved as a precise conditional theorem, theorem 5.3, needing all radii.
RH closure	*Not proved*. Requires an unbounded-slope family of valid QQRH certificates, or another independently sufficient global theorem.

7.3 Three realistic directions for extension

1. **Global radial-split certificates.** Exploit the already-derived trace $T_k(r)$ on arbitrary regular circles, with certified bounds for the complete residual zero contribution. No analytic continuation of the origin Taylor series is needed.
2. **Uniform radial sign estimates.** Prove $P_x(r, x) \geq 0$ on larger radii using canonical factors or a new integral kernel with a nonnegative remainder independent of unknown off-axis zeros.
3. **Moment inequalities with certified tails.** Establish coefficient inequalities such as (6.4) on systematically growing radial ranges, with quantitative error estimates strong enough to survive the infinite limit.

These are *research objectives*, not assertions established by the present trace identity.

Final statement. The original local Chebyshev identity $\pi r^k q_k$ extends to the global exact zero-aware trace $\pi T_k(r)$. The 255/7 anchor is a proved finite-radius sign region subject to its verified low-height input, and the height-to-radius theorem quantifies further *conditional* extensions. Full-angle angular positivity at *all* radii is equivalent to RH, while QQRH uses the corresponding sectorial sign condition. The new trace identities do not themselves prove that global sign.

CHAPTER A

The analytic height input in explicit constants

The supplied low-height Lean package derives (4.2) from Mathlib's pole-subtracted theta–Mellin representation. For clarity, the numerical constants in that intermediate estimate are reproduced here, without claiming a new independent formal check of the complete Mellin proof.

Write

$$p = \frac{6283}{2000}, \qquad q = \frac{100}{2313}, \qquad K(u) = \mathrm{hurwitzThetaMellinKernel}(u).$$

For $u \geq 1$, the archive supplies a theta-series envelope of the form

$$|K(u)| \leq \frac{2}{1-q^3}\, e^{-pu}. \tag{A.1}$$

The inversion formula for the kernel sends the range $0 < u \leq 1$ to the upper tail, retaining both endpoint weights u^{-1} and $u^{-1/2}$. The paired Mellin estimate is bounded by

$$\int_1^\infty \left(u^{-1} + u^{-1/2}\right) e^{-pu}\, \mathrm{d}u \leq \frac{e^{-p}}{p}\left(2 - \frac{1}{p+2} - \frac{1}{2p+3}\right). \tag{A.2}$$

The archive combines (A.1)–(A.2) with an explicit estimate for e^{-p} to obtain, for $0 \leq \mathrm{Re}\, s \leq 1$,

$$|\,\mathrm{completedRiemannZeta}_0(s)| \leq \frac{q}{p(1-q^3)}\left(2 - \frac{1}{p+2} - \frac{1}{2p+3}\right) \leq \frac{9}{385}. \tag{A.3}$$

This is the explicit input used in theorem 4.1. The following equalities are then plain algebra:

$$\left(\frac{13}{2}\right)^2 + \frac{1}{2} = \frac{171}{4}, \qquad \frac{171}{4} \cdot \frac{9}{385} = \frac{1539}{1540} < 1, \qquad \left(\frac{169}{4} + \frac{1}{4}\right)\frac{6}{7} = \frac{255}{7}.$$

The exact kernel definitions, integrability assumptions, and division conventions reside in the four archive modules `ThetaKernelAnchorBound`, `MellinAnchorBound`, `MellinPairedBound`, and `AnchorHeight`. None presupposes RH as a named premise in the reported theorem signatures.

CHAPTER B

Lean theorem names and verification boundary

The supplied QQRH Full-x Anchor 2 ZIP archive records the following theorem signatures in its `AUDIT.txt` and `README.md`:

- `AngularMaster.actual_xi_zero_thirteen_halves_lt_abs_im`: the height input $|\operatorname{Im}\rho| > 13/2$.
- `QQRH.second_full_x_radius`: the definition $R_* = 255/7$.
- `QQRH.second_full_x_monotone`: monotonicity of $P(r,\cdot)$ on $[-1,1]$ for $0 < r \le 255/7$.
- `QQRH.second_full_x_range_anchor`: the interior derivative nonnegativity conclusion.
- `QQRH.second_full_x_positive`: strict positivity of the actual profile on the closed angular interval and $0 \le r \le 255/7$.
- `QQRH.of_positivity`: the all-radius angular-positivity implication QQRH(m).

The present Chebyshev trace theorem is *derived in this chapter*; the cited archive is not asserted to contain its full Lean formalization. Its README reports Lean 4.33.0-compatible source and standard foundational axiom dependencies. The source and audit files were read for this exposition, but the full project has *not* been recompiled here, so this chapter does not claim an additional independent kernel-checked build.

The associated canonical-product infrastructure is derived in part from the upstream ξ formalization repository (Alejandro Z. and collaborators). That project explicitly distinguishes proved equivalences from a proof of RH.

CHAPTER C

Exercises with compact solutions

Problems

1. **Orthogonality.** From (1.7), derive (1.8) without quoting a Chebyshev reference.
2. **First moment.** Show that the trace identity for $k = 1$ can be expressed entirely through $\xi(\frac{1}{2})$ and $\xi''(\frac{1}{2})$.
3. **Exact angular cutoff.** Prove $x - a_m = 2(m^2\delta^2 - \gamma^2)/[(1 + m^2)(\delta^2 + \gamma^2)]$.
4. **The paired-factor threshold.** Starting from (4.7), show that the first nonnegative threshold of q is $(g - h)/(g + h)$.
5. **The missing sign region.** For fixed m, determine the part of $(0, \infty) \times (-1, 1)$ not covered by the finite-radius full-angle sign theorem and the all-radius sign condition $x > a_m$.
6. **Radial inversion.** Prove (5.2) from $z = (\delta + i\gamma)^2$.
7. **Kink test.** Verify (3.11) for the factor $G(z) = 1 + z/a$.
8. **Multiplicity.** Show directly that a transformed zero of multiplicity $\nu = 2$ forces the sign of P_x to change, even though $P_{xx}(r_0, x_0) = 0$.
9. **Global moments.** Starting from (3.5), prove the global derivative formula for $r > a$ when $G(z) = 1 + z/a$.
10. **Sign versus individual moments.** In the one-critical-line-factor model, show that $q_2 < 0$ although $\partial_x \log P_G > 0$.

Solutions

1. Put $x = \cos\theta$ and use $\mathrm{d}x = -\sin\theta\,\mathrm{d}\theta$. The weighted product becomes $\int_0^\pi \sin((j+1)\theta)\sin((k+1)\theta)\,\mathrm{d}\theta = \frac{\pi}{2}\mathbf{1}_{\{j=k\}}$.
2. Equations (2.11) and (2.13) give $\displaystyle\int_{-1}^{1} \mathcal{L}(r, x)\sqrt{1 - x^2}\,\mathrm{d}x = \frac{\pi r\,\xi''(\frac{1}{2})}{2\xi(\frac{1}{2})}$.
3. Expand and simplify the numerator after cross-multiplying the two positive denominators; the δ^2 terms give $2m^2\delta^2$ and the γ^2 terms give $-2\gamma^2$.
4. Solve $(-c)(1 + q^2) - 2q = 0$ with $-c = (g^2 - h^2)/(g^2 + h^2)$, using $\sqrt{1 - c^2} = 2gh/(g^2 + h^2)$; the smaller root is $(g - h)/(g + h)$.
5. The missing set is precisely $\{(r, x) : r > 255/7,\ -1 < x \le a_m\}$.
6. One zero $-a$ yields $q_k = (-1)^{k+1}a^{-k}$, so $q_2 = -a^{-2}$. The angular derivative is nevertheless $2(r/a)/(1 + 2(r/a)x + (r/a)^2) > 0$ wherever defined.
7. $r = \delta^2 + \gamma^2$ and $rx = \delta^2 - \gamma^2$. Add and subtract to get the two square-root relations.

8. For $\alpha = -a$, the contribution to T_k is $(-1)^{k+1}(r/a)^k$ below a and $(-1)^{k+1}(a/r)^k$ above a. The derivative jump at a is $2k(-1)^k/a$, matching (3.11).

9. $P(r_0, x) = C(x - x_0)^4(1 + O(x - x_0))$, so $P_x(r_0, x) = 4C(x - x_0)^3(1 + O(x - x_0))$, which is negative immediately to the left.

10. With $u = a/r < 1$ and $P_G(r, x) = (r/a)^2(1 + 2ux + u^2)$, differentiate its logarithm to get $2u/(1 + 2ux + u^2) = 2\sum_{k\geq 1}(-1)^{k+1}u^k U_{k-1}(x)$.

References and source note

[1] NIST Digital Library of Mathematical Functions, § 25.4, *Reflection Formulas*, https://dlmf.nist.gov/25.4; and § 25.10, *Zeros*, https://dlmf.nist.gov/25.10.

[2] E. C. Titchmarsh, revised by D. R. Heath-Brown, *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford University Press, 1986. Standard theory of the completed zeta function and its zero divisor.

[3] G. Szegő, *Orthogonal Polynomials*, AMS Colloquium Publications, Vol. 23. Standard orthogonality of Chebyshev polynomials.

[4] B. Ya. Levin, *Distribution of Zeros of Entire Functions*, American Mathematical Society. Canonical products, Jensen's formula, and logarithmic potentials.

[5] *QQRH Full-x Anchor 2*, Lean 4.33.0 project archive, `QQRH_FULL_X_ANCHOR_2_LEAN_v0_1_0.zip`.

Relevant modules: the theta–Mellin height bound, full-angle factor geometry, canonical-product limit, and second anchor theorem.

[6] RiemannHypothesis-Formalization, upstream Lean/Mathlib canonical-product and xi formalization reference (GitHub, Alejandro Z.).

Prepared 20 October 2026 (second edition). Mathematical status: exact local and global radial-split Chebyshev traces, analytic RH positivity equivalence, and finite angular anchor with explicit low-height input. The universal sign requirement remains unproved here; no proof of RH is claimed.